Week 6, Lecture 3
Quantum Optical Metrology

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Office hours: **This Week Thursday 1-2 pm**

Bachor & Ralph, Chapters 10.3 and 10.4

Reminder

Office hours will be held from 1.00 to 2.00 pm this Thursday (note change from previous weeks)

Assessment

Reminder that assignment 2 is due 04/09/26 at 4pm

Seminar: coming up in Week 10 and 11

Exam: The best way to prepare for the quantum optics part of the exam is with the tutorial questions, assignments and past exams.

Thanks for exploring quantum optics with me!

This lecture concludes both the quantum optics section and my part of the course. If you have any questions about anything we have covered, please don't hesitate to email me.

Recap: The Jaynes–Cummings model

One two-level atom interacts with one quantised cavity mode:

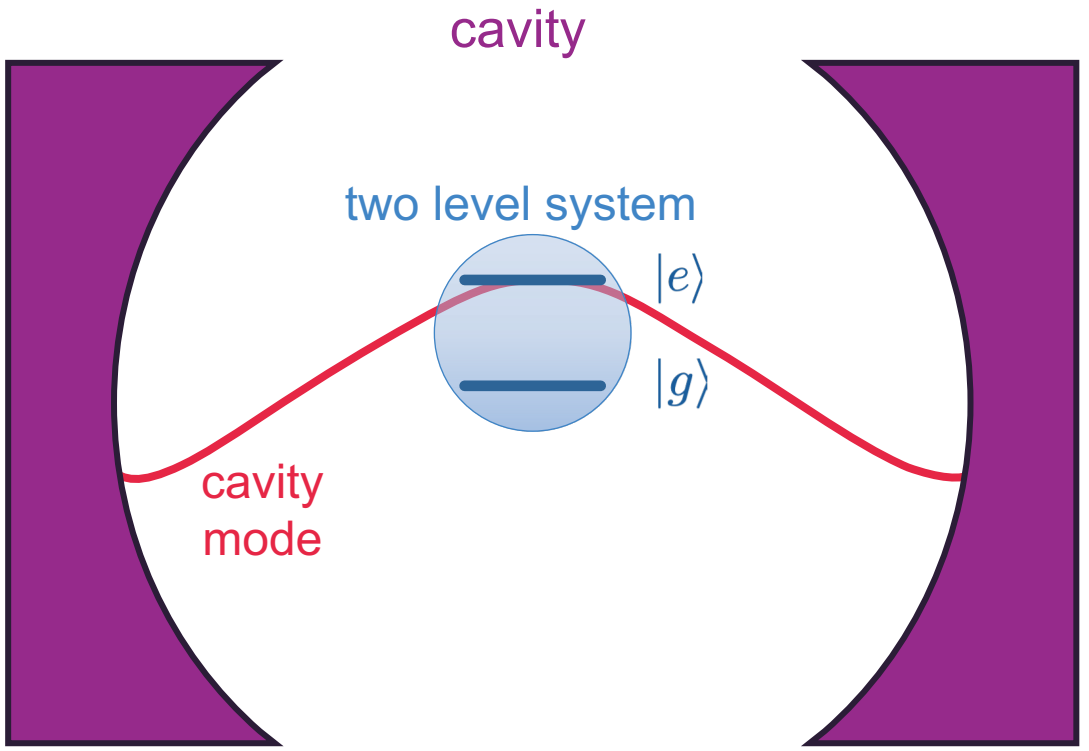
$$H_{\mathrm{JC}} = \hbar\omega_c a^\dagger a + \hbar\omega_a \sigma_+\sigma_- + \hbar g\left(a\sigma_+ + a^\dagger\sigma_-\right)$$

The interaction exchanges an excitation between the atom and cavity:

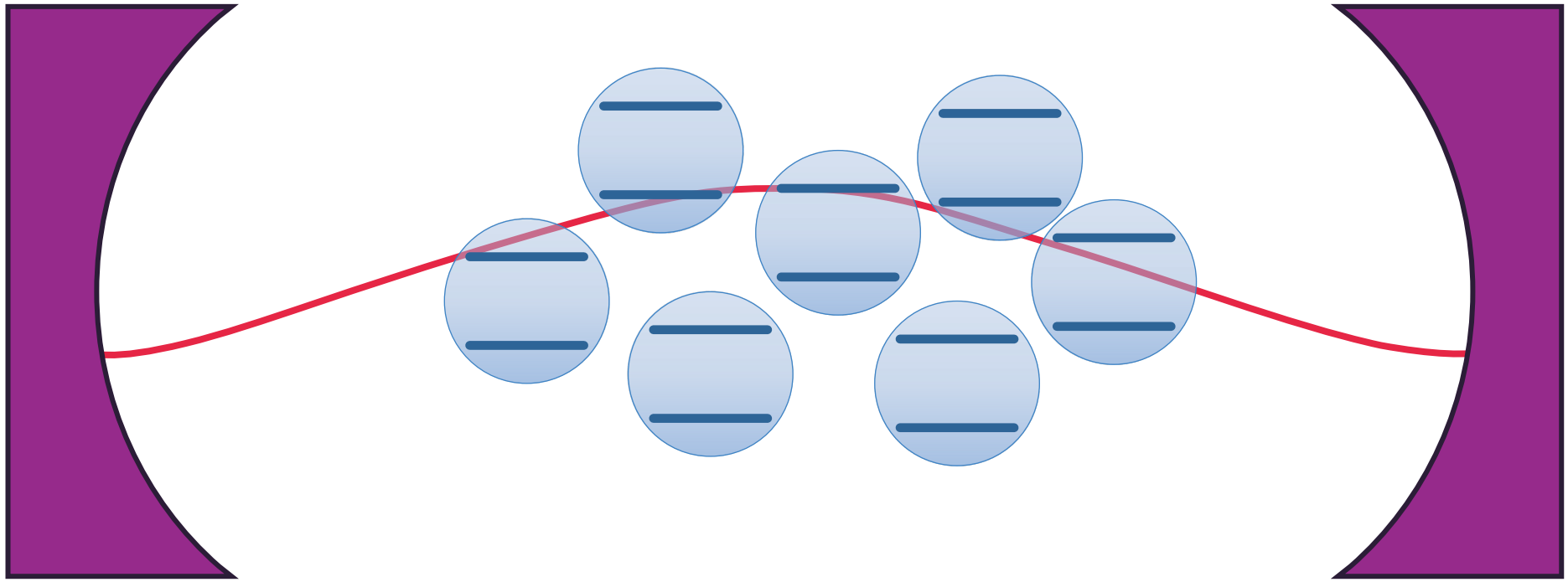
$$|e,n\rangle \longleftrightarrow |g,n+1\rangle$$

The total number of excitations is conserved:

$$\hat{N}_{\mathrm{exc}} = a^\dagger a + \sigma_+\sigma_-$$



Recap: The Tavis–Cummings model and superradiance



What happens when N atoms all couple to the same cavity mode?

Quantum Optical Metrology

Quantum limits, interferometry and squeezed-light sensing



What you will learn today

The University Of Queensland
AUSTRALIA

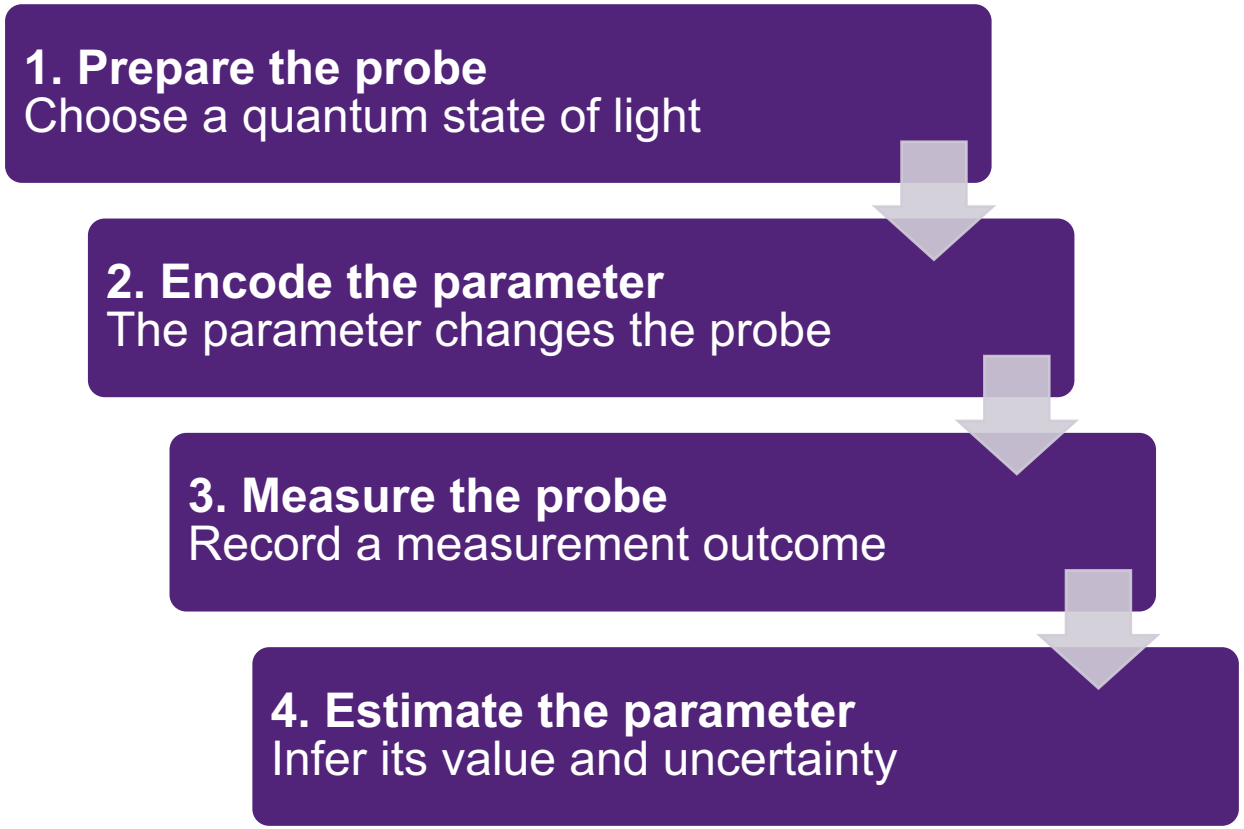
How to describe parameter estimation using quantum states and measurements

How to derive the quantum limit for optical phase estimation

Why squeezed light improves the sensitivity of real sensors

What is quantum metrology?

Quantum metrology uses quantum systems as probes to estimate physical parameters as precisely as possible.



Describing an estimate

An estimate has both a value and an uncertainty

θ: the unknown parameter

$\tilde{\theta}$: an estimate obtained from the measurement data

M: the number of independent measurements

An unbiased estimator satisfies

$$\left\langle \tilde{\theta} \right\rangle = \theta$$

Its precision is described by the variance

$$\left(\Delta\tilde{\theta}\right)^2 = \left\langle \left(\tilde{\theta} - \theta\right)^2 \right\rangle$$

Smaller variance means a more precise estimate.

Encoding the parameter

The parameter is encoded in the probe state

Begin with a known probe state ρ_0.

The unknown parameter θ produces the unitary transformation

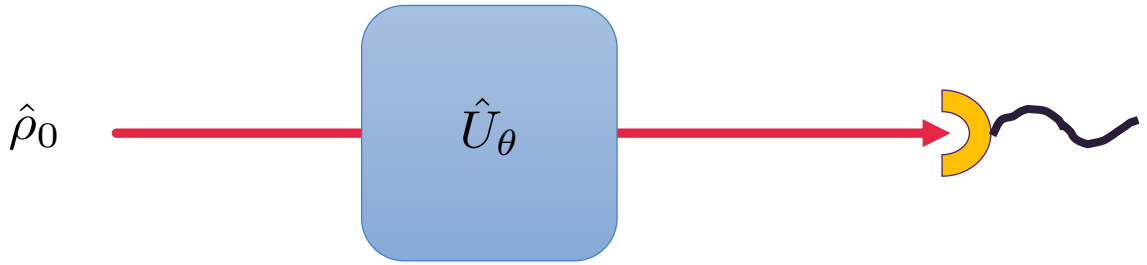
$$\hat{U}_\theta = e^{-i\theta\hat{G}}$$

The resulting state is

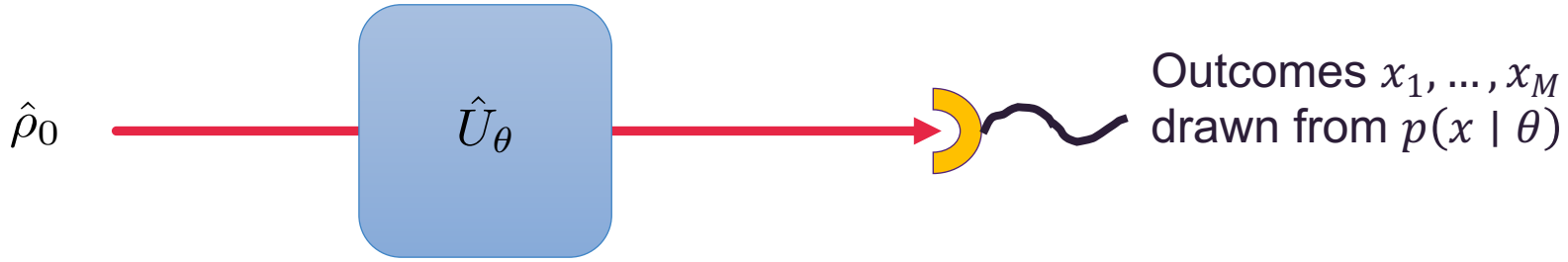
$$\rho_\theta = \hat{U}_\theta \rho_0 \hat{U}_\theta^\dagger$$

$\hat{G}$ is the generator of the change.

Different values of θ produce different output states.



From measurement outcomes to an estimate



The Cramér–Rao inequality gives

$$\left(\Delta\tilde{\theta}\right)^2 \geq \frac{1}{MF(\theta)}$$

variance of estimated parameter

number of measurements

$F(\theta)$: Fisher information contained in the measurement outcomes

Classical Fisher information

For a chosen measurement with outcomes distributed according to $p(x \mid \theta)$,

$$F_C(\theta) = \int dx\, p(x|\theta) \left[\frac{\partial}{\partial \theta} \ln p(x|\theta) \right]^2$$

- Measures how strongly the outcome probabilities depend on θ
- Depends on the chosen measurement
- Larger Fisher information allows a more precise estimate

Quantum Fisher information

The quantum Fisher information is the largest Fisher information available from the encoded state:

A chosen quantum measurement $\{\hat{\Pi}_x\}$produces the outcome probabilities

$$p(x|\theta) = \mathrm{Tr}\left(\rho_\theta \hat{\Pi}_x\right)$$

Different measurements generally produce different Fisher information.

The quantum Fisher information is the maximum over all possible measurements:

$$F_Q(\theta) = \max_{\{\hat{\Pi}_x\}} F(\theta)$$

Therefore,

$$F(\theta) \leq F_Q(\theta)$$

Large F_Q means that nearby parameter values produce more distinguishable quantum states.

The quantum Cramér–Rao bound

For any chosen measurement,

$$\left(\Delta\tilde{\theta}\right)^2 \geq \frac{1}{MF_C(\theta)} \geq \frac{1}{MF_Q(\theta)}$$

Therefore, the **quantum Cramér–Rao bound** is

$$\left(\Delta\tilde{\theta}\right)^2 \geq \frac{1}{MF_Q(\theta)}$$

For a pure probe undergoing unitary encoding generated by $\hat{G}$

$$F_Q = 4\left(\Delta G\right)^2 = 4\left(\left\langle \hat{G}^2 \right\rangle - \left\langle \hat{G} \right\rangle^2\right)$$

Hence,

$$\left(\Delta\tilde{\theta}\right)^2 \geq \frac{1}{4M\left(\Delta G\right)^2}$$

Quantum limit to phase estimation

Photon number generates an optical phase shift

An unknown optical phase ϕ is encoded through

$$\hat{U}_\phi = e^{-i\phi\hat{n}}, \qquad \hat{n} = \hat{a}^\dagger \hat{a}$$

For a pure probe state,

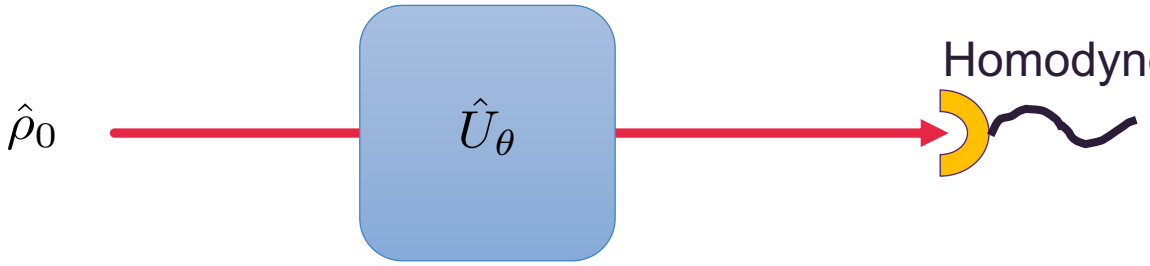
$$F_Q = 4\,(\Delta n)^2$$

The quantum Cramér–Rao bound becomes

$$\left(\Delta\tilde{\phi}\right)^2 \geq \frac{1}{4M\,(\Delta n)^2}$$

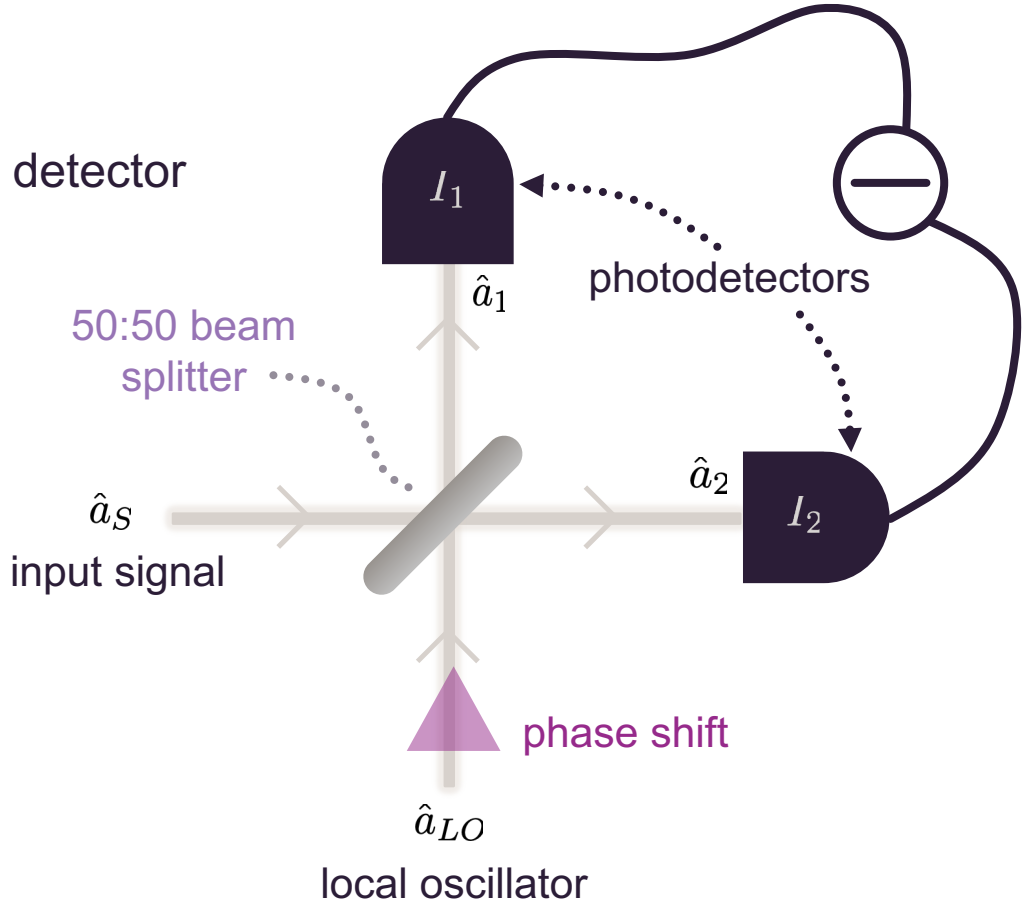
Phase sensitivity is determined by the photon-number variance of the probe.

Homodyne detection



- The local oscillator provides a phase reference.
- A small phase shift becomes a change in the measured photocurrent.
- Ideal homodyne detection can saturate this bound

$$\left(\Delta\tilde{\phi}\right)^2 \geq \frac{1}{4M\left(\Delta n\right)^2}$$



Coherent-state phase estimation

Coherent light – photon number variance

- **Photon number variance**

$$\mathrm{var}\,(n) = \langle \Delta \hat{n}^2 \rangle = \langle \hat{n}^2 \rangle - \langle \hat{n} \rangle^2$$

$$\langle \hat{n}^2 \rangle = \langle \alpha | \hat{a}^\dagger \hat{a} \hat{a}^\dagger \hat{a} | \alpha \rangle$$

$$\langle \hat{n}^2 \rangle = |\alpha|^2 \langle \alpha | \hat{a} \hat{a}^\dagger | \alpha \rangle$$

$$\hat{a} |\alpha\rangle = \alpha |\alpha\rangle$$

$$\langle \alpha | \hat{a}^\dagger = \alpha^* \langle \alpha |$$

$$[\hat{a}, \hat{a}^\dagger] = 1 \qquad \hat{a}\hat{a}^\dagger = 1 + \hat{a}^\dagger \hat{a}$$

$$\langle \hat{n}^2 \rangle = |\alpha|^2 \langle \alpha | (1 + \hat{a}^\dagger \hat{a}) | \alpha \rangle$$

$$\langle \hat{n}^2 \rangle = |\alpha|^2 (1 + |\alpha|^2)$$

$$\langle \hat{n}^2 \rangle = |\alpha|^4 + |\alpha|^2$$

$$\mathrm{var}\,(n) = \langle \hat{n}^2 \rangle - \langle \hat{n} \rangle^2$$

$$\mathrm{var}\,(n) = |\alpha|^4 + |\alpha|^2 - (|\alpha|^2)^2$$

$$\mathrm{var}\,(n) = |\alpha|^2 = \bar{n}$$

Coherent-state phase estimation

A coherent state gives shot-noise-limited sensitivity

For a coherent state $|\alpha\rangle$

$$\bar{n} = \langle \hat{n} \rangle = |\alpha|^2, \qquad (\Delta n)^2 = \bar{n}$$

Therefore,

$$F_Q = 4\bar{n}$$

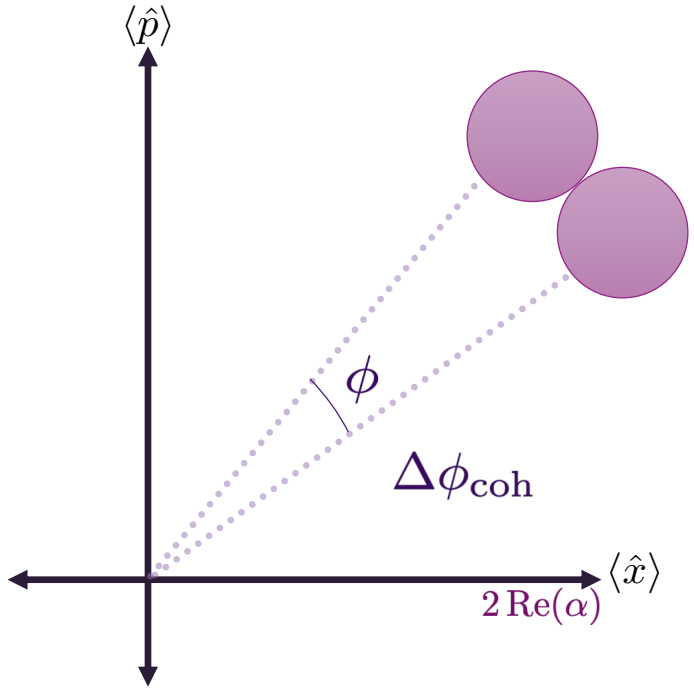
The phase-estimation bound becomes

$$\left(\Delta\tilde{\phi}\right)^2 \geq \frac{1}{4M\bar{n}}, \qquad \Delta\tilde{\phi} \geq \frac{1}{2\sqrt{M\bar{n}}}$$

Shot-noise scaling:

$$\Delta\tilde{\phi} \propto \frac{1}{\sqrt{\bar{n}}}$$

Standard Quantum limit



Coherent state

Displaced squeezed state

Assume

$$|\psi\rangle = \hat{D}(\alpha)\hat{S}(r)|0\rangle, \qquad \hat{S}^\dagger(r)\hat{a}\hat{S}(r) = \hat{a}\cosh r + \hat{a}^\dagger \sinh r \qquad \hat{n} = \hat{a}^\dagger\hat{a} \qquad \hat{n}^2 = \hat{a}^\dagger\hat{a}\,\hat{a}^\dagger\hat{a}$$

The variance is

$$(\Delta n)^2 = \langle\psi|\hat{n}^2|\psi\rangle - \langle\psi|\hat{n}|\psi\rangle^2$$

Move the displacement and squeezing onto the operators:

$$\hat{S}^\dagger\hat{D}^\dagger\hat{a}\hat{D}\hat{S} = \alpha + \cosh r\,\hat{a} + \sinh r\,\hat{a}^\dagger$$
$$\hat{S}^\dagger\hat{D}^\dagger\hat{a}^\dagger\hat{D}\hat{S} = \alpha^* + \cosh r\,\hat{a}^\dagger + \sinh r\,\hat{a}$$

$$\langle\hat{n}\rangle = |\alpha|^2 + \sinh^2 r$$

Taking α to be real, write the variance as

$$\begin{aligned}(\Delta n)^2 = \Big\langle &(\alpha + \cosh r\,\hat{a}^\dagger + \sinh r\,\hat{a})(\alpha + \cosh r\,\hat{a} + \sinh r\,\hat{a}^\dagger) \\ &\times (\alpha + \cosh r\,\hat{a}^\dagger + \sinh r\,\hat{a})(\alpha + \cosh r\,\hat{a} + \sinh r\,\hat{a}^\dagger)\Big\rangle_0 \\ &- \left(|\alpha|^2 + \sinh^2 r\right)^2 .\end{aligned}$$

Displaced squeezed state

$$(\Delta n)^2 = \Big\langle (\alpha + \cosh r\,\hat{a}^\dagger + \sinh r\,\hat{a})(\alpha + \cosh r\,\hat{a} + \sinh r\,\hat{a}^\dagger) \times (\alpha + \cosh r\,\hat{a}^\dagger + \sinh r\,\hat{a})(\alpha + \cosh r\,\hat{a} + \sinh r\,\hat{a}^\dagger) \Big\rangle_0 - \left(|\alpha|^2 + \sinh^2 r\right)^2 .$$

Expanding the vacuum expectation value gives

$$\langle \hat{n}^2 \rangle = |\alpha|^4 + |\alpha|^2 \left(\cosh^2 r + 3\sinh^2 r + 2\sinh r \cosh r\right) + 3\sinh^4 r + 2\sinh^2 r$$

Therefore,

$$(\Delta n)^2 = |\alpha|^2 (\cosh r + \sinh r)^2 + 2\sinh^2 r \cosh^2 r$$

For $|r| \gg 1, \qquad \sinh^2 |r| \simeq |\alpha|^2, \qquad \bar{n} \simeq 2|\alpha|^2$

We have

$$(\Delta n)^2 \simeq 2|\alpha|^4 \simeq \frac{\bar{n}^2}{2}$$

Displaced squeezed state

$$(\Delta n)^2 \simeq 2|\alpha|^4 \simeq \frac{\bar{n}^2}{2}$$

Displaced squeezed state

Consequently, for one probe,

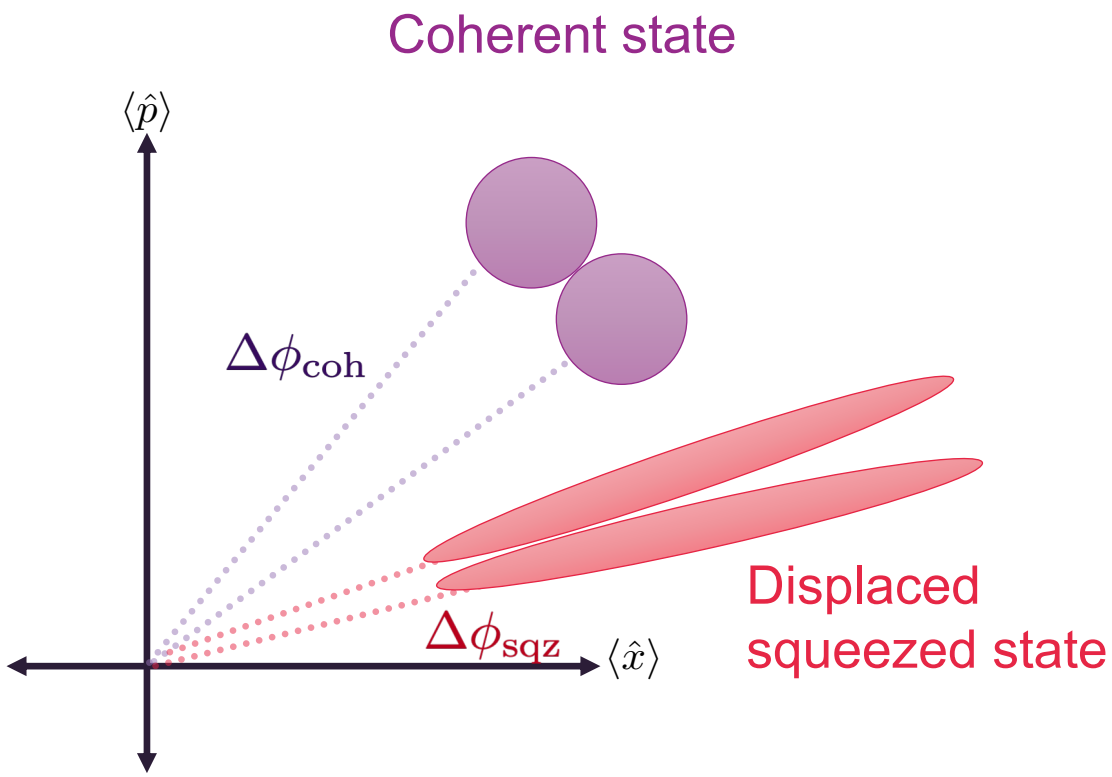
$$(\Delta\tilde{\phi})^2 \geq \frac{1}{4(\Delta n)^2} \simeq \frac{1}{2\bar{n}^2}$$

$$\Delta\phi \propto \frac{1}{\bar{n}}$$ **Heisenberg limit**

Coherent states

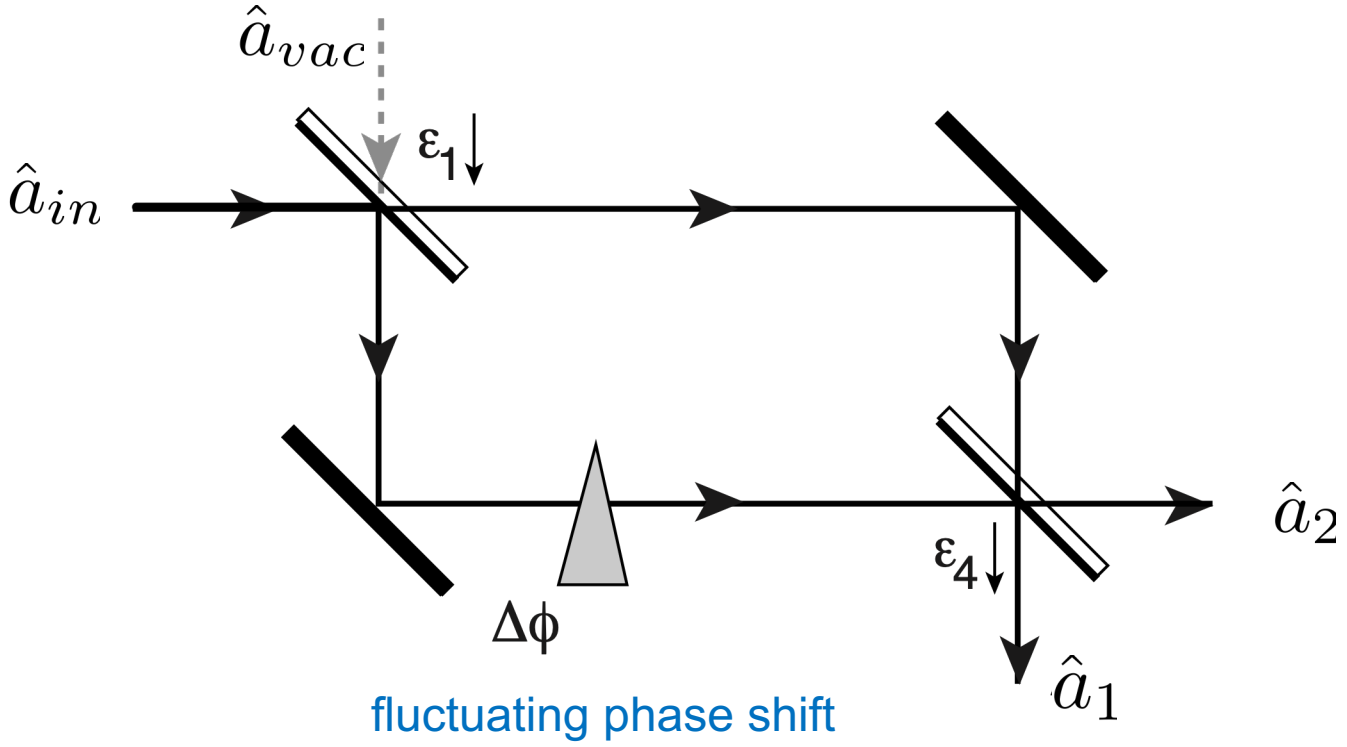
Shot-noise scaling:

$$\Delta\tilde{\phi} \propto \frac{1}{\sqrt{\bar{n}}}$$ **Standard Quantum limit**



Mach–Zehnder interferometer

- The first beam splitter divides the optical field between two paths.
- The paths acquire a relative phase shift.
- The second beam splitter converts this phase difference into a change in the output intensities.
- The intensity difference between the detectors provides the measurement signal.



Mach–Zehnder interferometer

$$\hat{b}_{\text{upper}} = \frac{1}{\sqrt{2}}(\hat{a}_{\text{in}} - \hat{a}_{\text{vac}})$$

$$\hat{b}_{\text{lower}} = \frac{1}{\sqrt{2}}(\hat{a}_{\text{in}} + \hat{a}_{\text{vac}})$$

The phase shift then acts on the lower arm:

$$\hat{b}'_{\text{upper}} = \hat{b}_{\text{upper}}, \qquad \hat{b}'_{\text{lower}} = e^{i\Delta\phi}\hat{b}_{\text{lower}}$$

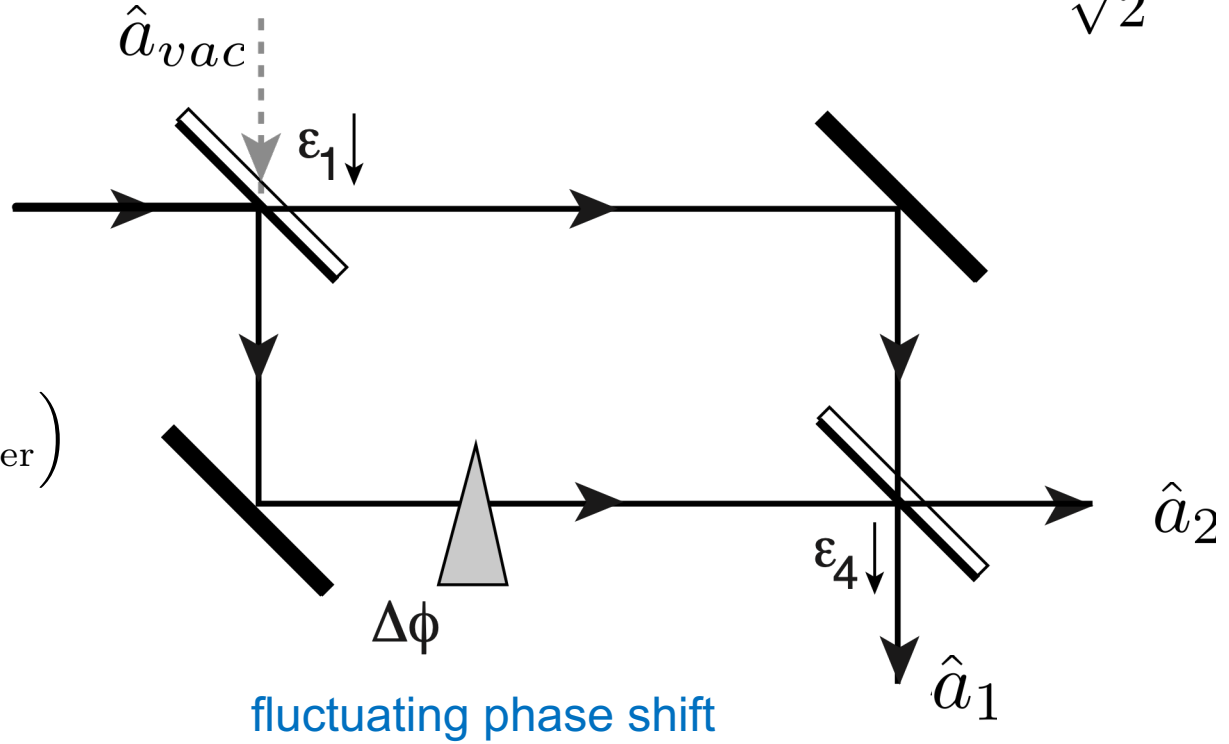
At the second 50:50 beam splitter,

$$\hat{a}_1 = \frac{1}{\sqrt{2}}\left(\hat{b}'_{\text{upper}} + \hat{b}'_{\text{lower}}\right) \quad \hat{a}_2 = \frac{1}{\sqrt{2}}\left(-\hat{b}'_{\text{upper}} + \hat{b}'_{\text{lower}}\right)$$

$$\hat{a}_1 = \frac{1}{\sqrt{2}}\left(\hat{b}_{\text{upper}} + e^{i\Delta\phi}\hat{b}_{\text{lower}}\right)$$

$$\hat{a}_2 = \frac{1}{\sqrt{2}}\left(-\hat{b}_{\text{upper}} + e^{i\Delta\phi}\hat{b}_{\text{lower}}\right)$$

$$\varepsilon_1 = \varepsilon_4 = \frac{1}{\sqrt{2}}$$



fluctuating phase shift

Mach–Zehnder interferometer

$$\hat{a}_1 = \frac{1}{\sqrt{2}}\left[\frac{1}{\sqrt{2}}\left(\hat{a}_{\mathrm{in}} - \hat{a}_{\mathrm{vac}}\right) + e^{i\Delta\phi}\frac{1}{\sqrt{2}}\left(\hat{a}_{\mathrm{in}} + \hat{a}_{\mathrm{vac}}\right)\right]$$

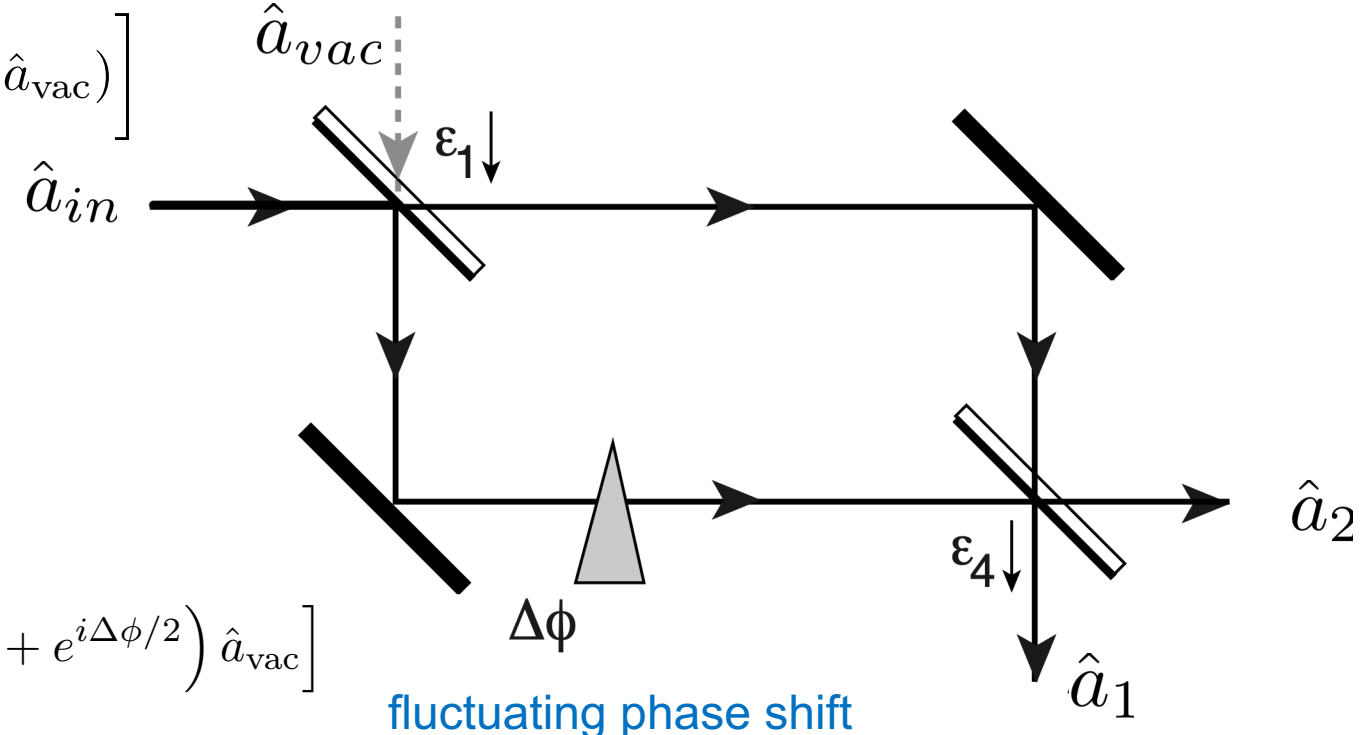
$$\hat{a}_2 = \frac{1}{\sqrt{2}}\left[-\frac{1}{\sqrt{2}}\left(\hat{a}_{\mathrm{in}} - \hat{a}_{\mathrm{vac}}\right) + e^{i\Delta\phi}\frac{1}{\sqrt{2}}\left(\hat{a}_{\mathrm{in}} + \hat{a}_{\mathrm{vac}}\right)\right]$$

$$\hat{a}_1 = \frac{1}{2}\left[\left(1 + e^{i\Delta\phi}\right)\hat{a}_{\mathrm{in}} + \left(-1 + e^{i\Delta\phi}\right)\hat{a}_{\mathrm{vac}}\right]$$

$$\hat{a}_2 = \frac{1}{2}\left[\left(-1 + e^{i\Delta\phi}\right)\hat{a}_{\mathrm{in}} + \left(1 + e^{i\Delta\phi}\right)\hat{a}_{\mathrm{vac}}\right]$$

$$\hat{a}_1 = \frac{e^{i\Delta\phi/2}}{2}\left[\left(e^{-i\Delta\phi/2} + e^{i\Delta\phi/2}\right)\hat{a}_{\mathrm{in}} + \left(-e^{-i\Delta\phi/2} + e^{i\Delta\phi/2}\right)\hat{a}_{\mathrm{vac}}\right]$$

$$\hat{a}_2 = \frac{e^{i\Delta\phi/2}}{2}\left[\left(-e^{-i\Delta\phi/2} + e^{i\Delta\phi/2}\right)\hat{a}_{\mathrm{in}} + \left(e^{-i\Delta\phi/2} + e^{i\Delta\phi/2}\right)\hat{a}_{\mathrm{vac}}\right]$$



fluctuating phase shift

Mach–Zehnder interferometer

$$\hat{a}_1 = \frac{e^{i\Delta\phi/2}}{2}\left[\left(e^{-i\Delta\phi/2} + e^{i\Delta\phi/2}\right)\hat{a}_{\text{in}} + \left(-e^{-i\Delta\phi/2} + e^{i\Delta\phi/2}\right)\hat{a}_{\text{vac}}\right]$$

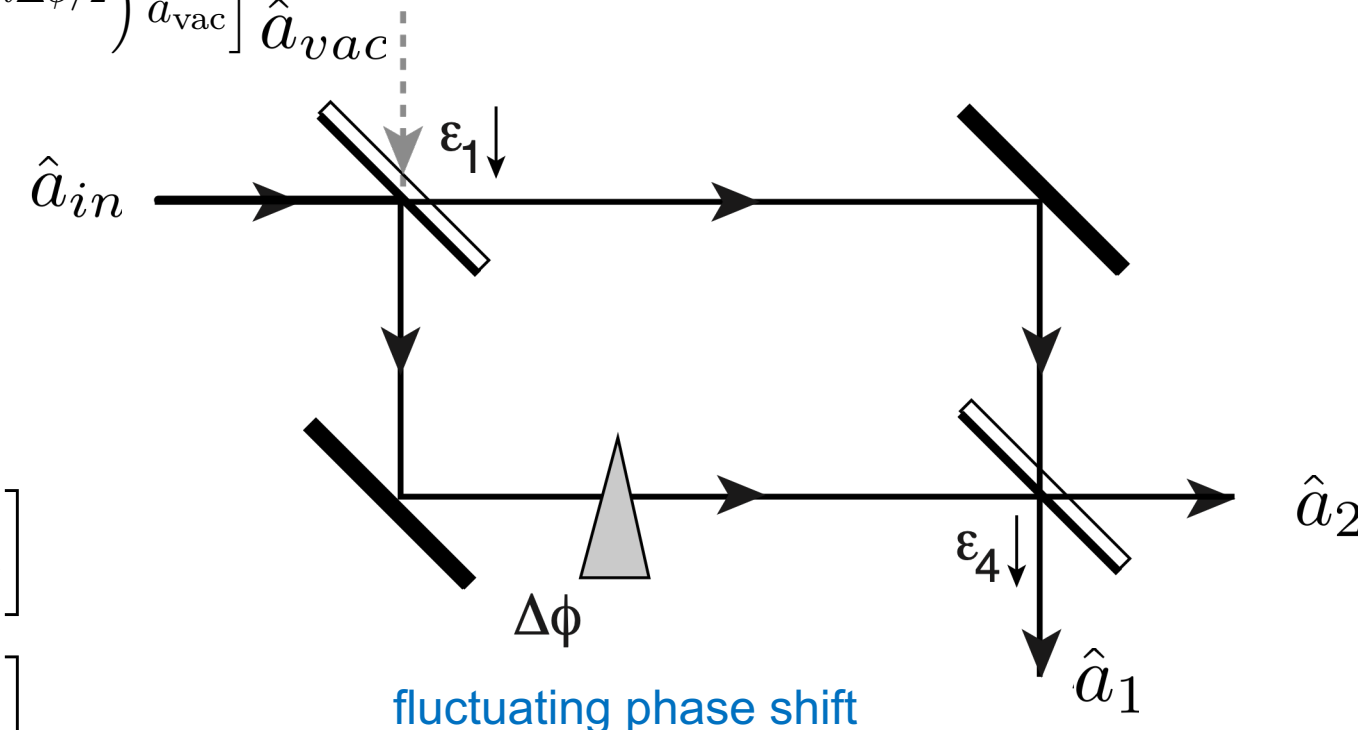
$$\hat{a}_2 = \frac{e^{i\Delta\phi/2}}{2}\left[\left(-e^{-i\Delta\phi/2} + e^{i\Delta\phi/2}\right)\hat{a}_{\text{in}} + \left(e^{-i\Delta\phi/2} + e^{i\Delta\phi/2}\right)\hat{a}_{\text{vac}}\right]$$

$$e^{-i\Delta\phi/2} + e^{i\Delta\phi/2} = 2\cos\left(\frac{\Delta\phi}{2}\right)$$

$$-e^{-i\Delta\phi/2} + e^{i\Delta\phi/2} = 2i\sin\left(\frac{\Delta\phi}{2}\right)$$

$$\hat{a}_1 = e^{i\Delta\phi/2}\left[\cos\left(\frac{\Delta\phi}{2}\right)\hat{a}_{\text{in}} + i\sin\left(\frac{\Delta\phi}{2}\right)\hat{a}_{\text{vac}}\right]$$

$$\hat{a}_2 = e^{i\Delta\phi/2}\left[\cos\left(\frac{\Delta\phi}{2}\right)\hat{a}_{\text{vac}} + i\sin\left(\frac{\Delta\phi}{2}\right)\hat{a}_{\text{in}}\right]$$



fluctuating phase shift

Mach–Zehnder interferometer

$$\hat{a}_1 = e^{i\Delta\phi/2}\left[\cos\left(\frac{\Delta\phi}{2}\right)\hat{a}_{\text{in}} + i\sin\left(\frac{\Delta\phi}{2}\right)\hat{a}_{\text{vac}}\right]$$

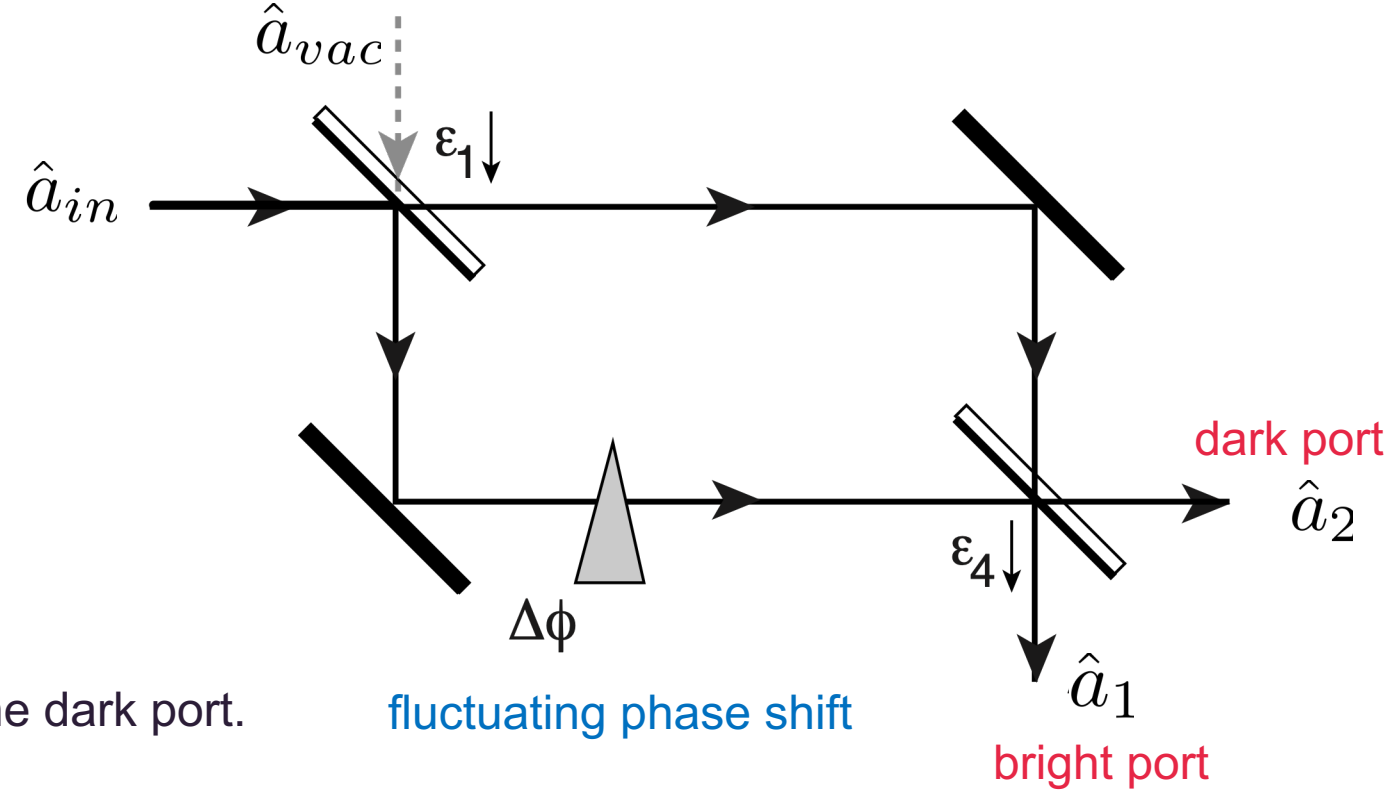
$$\hat{a}_2 = e^{i\Delta\phi/2}\left[\cos\left(\frac{\Delta\phi}{2}\right)\hat{a}_{\text{vac}} + i\sin\left(\frac{\Delta\phi}{2}\right)\hat{a}_{\text{in}}\right]$$

Now if $\phi = 0$:

$$\hat{a}_1 = e^0\left[\cos(0)\hat{a}_{\text{in}} + i\sin(0)\hat{a}_{\text{vac}}\right] = \hat{a}_{\text{in}}$$

$$\hat{a}_2 = e^0\left[i\sin(0)\hat{a}_{\text{in}} + \cos(0)\hat{a}_{\text{vac}}\right] = \hat{a}_{\text{vac}}$$

Thus, output 1 is the bright port and output 2 is the dark port.



Mach–Zehnder interferometer

Start from the dark-port output:

$$\hat{a}_2 = e^{i\Delta\phi/2}\left[\cos\left(\frac{\Delta\phi}{2}\right)\hat{a}_{\mathrm{vac}} + i\sin\left(\frac{\Delta\phi}{2}\right)\hat{a}_{\mathrm{in}}\right]$$

Work in the differential-phase frame:

$$\widetilde{\hat{a}}_2 \equiv e^{-i\Delta\phi/2}\hat{a}_2 = i\sin\left(\frac{\Delta\phi}{2}\right)\hat{a}_{\mathrm{in}} + \cos\left(\frac{\Delta\phi}{2}\right)\hat{a}_{\mathrm{vac}}$$

$|\Delta\phi| \ll 1$:

$$\sin\left(\frac{\Delta\phi}{2}\right) \simeq \frac{\Delta\phi}{2}, \qquad \cos\left(\frac{\Delta\phi}{2}\right) \simeq 1$$

Therefore,

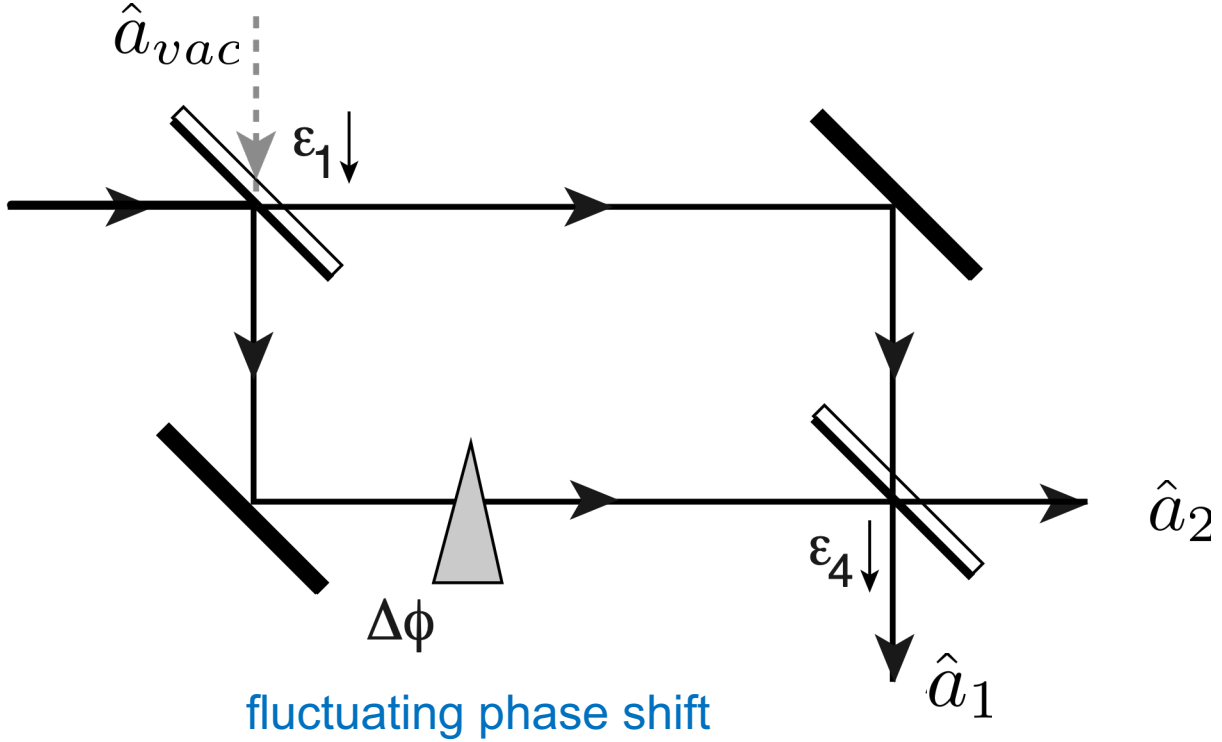
$$\widetilde{\hat{a}}_2 \simeq i\frac{\Delta\phi}{2}\hat{a}_{\mathrm{in}} + \hat{a}_{\mathrm{vac}}$$

Take a strong coherent input

$$\hat{a}_{\mathrm{in}} = \alpha + \delta\hat{a}_{\mathrm{in}}, \qquad \alpha = \sqrt{\bar{n}} \in \mathbb{R}, \qquad \alpha \gg 1$$

Substituting gives

$$\widetilde{\hat{a}}_2 \simeq \hat{a}_{\mathrm{vac}} + i\frac{\alpha\Delta\phi}{2} + i\frac{\Delta\phi}{2}\delta\hat{a}_{\mathrm{in}}$$



fluctuating phase shift

Mach–Zehnder interferometer

$$\tilde{\hat{a}}_2 \simeq \hat{a}_{\rm vac} + i\frac{\alpha\Delta\phi}{2} + i\frac{\Delta\phi}{2}\delta\hat{a}_{\rm in}$$

Keep the leading terms

$$\tilde{\hat{a}}_2 \simeq \hat{a}_{\rm vac} + i\frac{\sqrt{\bar{n}}\,\Delta\phi}{2}$$

Form the phase quadrature:

$$\tilde{\hat{X}}_2^- \equiv i\left(\tilde{\hat{a}}_2^\dagger - \tilde{\hat{a}}_2\right)$$

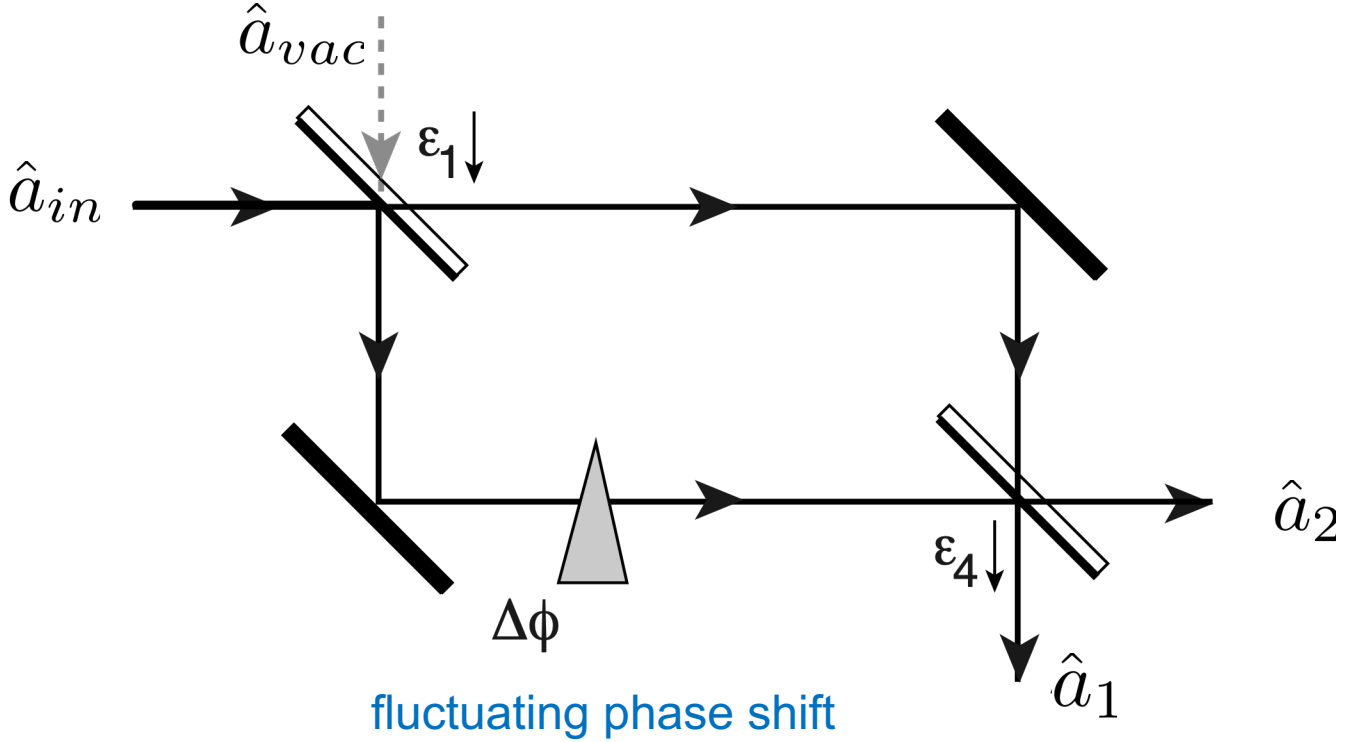
Since

$$\tilde{\hat{a}}_2^\dagger \simeq \hat{a}_{\rm vac}^\dagger - i\frac{\sqrt{\bar{n}}\,\Delta\phi}{2}$$

we obtain $\tilde{\hat{X}}_2^- \simeq i\left[\hat{a}_{\rm vac}^\dagger - i\frac{\sqrt{\bar{n}}\Delta\phi}{2} - \hat{a}_{\rm vac} - i\frac{\sqrt{\bar{n}}\Delta\phi}{2}\right]$

$$= i\left(\hat{a}_{\rm vac}^\dagger - \hat{a}_{\rm vac}\right) + \sqrt{\bar{n}}\,\Delta\phi$$

$$= \boxed{\underbrace{\hat{X}_{\rm vac}^-}_{\text{quantum noise}} + \underbrace{\sqrt{\bar{n}}\,\Delta\phi}_{\text{phase signal}}}$$



fluctuating phase shift

Mach–Zehnder interferometer

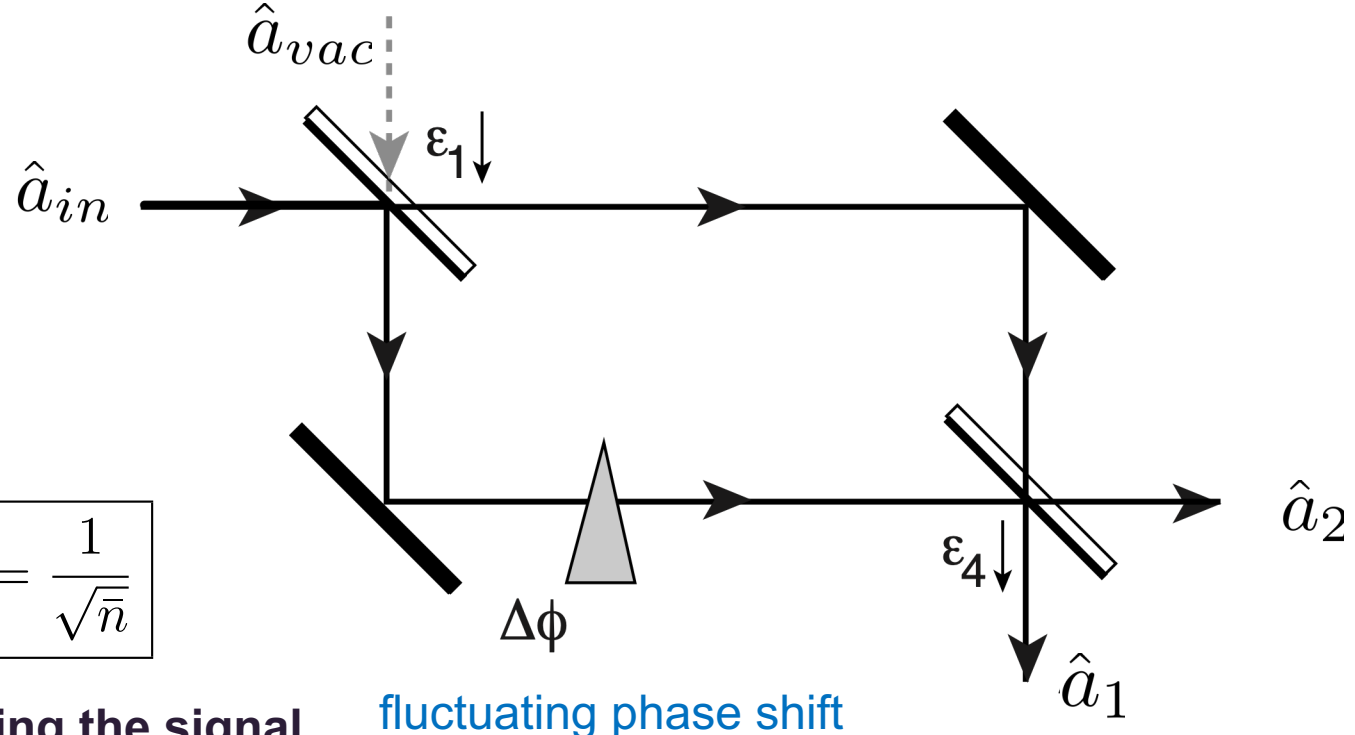
$$\tilde{\hat{X}}_2^- \simeq = \boxed{\underbrace{\hat{X}^-_{\rm vac}}_{\text{quantum noise}} + \underbrace{\sqrt{\bar{n}}\,\Delta\phi}_{\text{phase signal}}}$$

Signal power: $S \equiv \left|\left\langle \hat{X}_2^- \right\rangle\right|^2 = \bar{n}(\Delta\phi)^2$

Noise power: $N \equiv \left(\Delta X_2^-\right)^2 = V^-_{\rm vac} = 1$

Therefore

$$\frac{S}{N} = \bar{n}(\Delta\phi)^2, \qquad \frac{S}{N} = 1 \implies \boxed{\Delta\phi_{\rm SQL} = \frac{1}{\sqrt{\bar{n}}}}$$

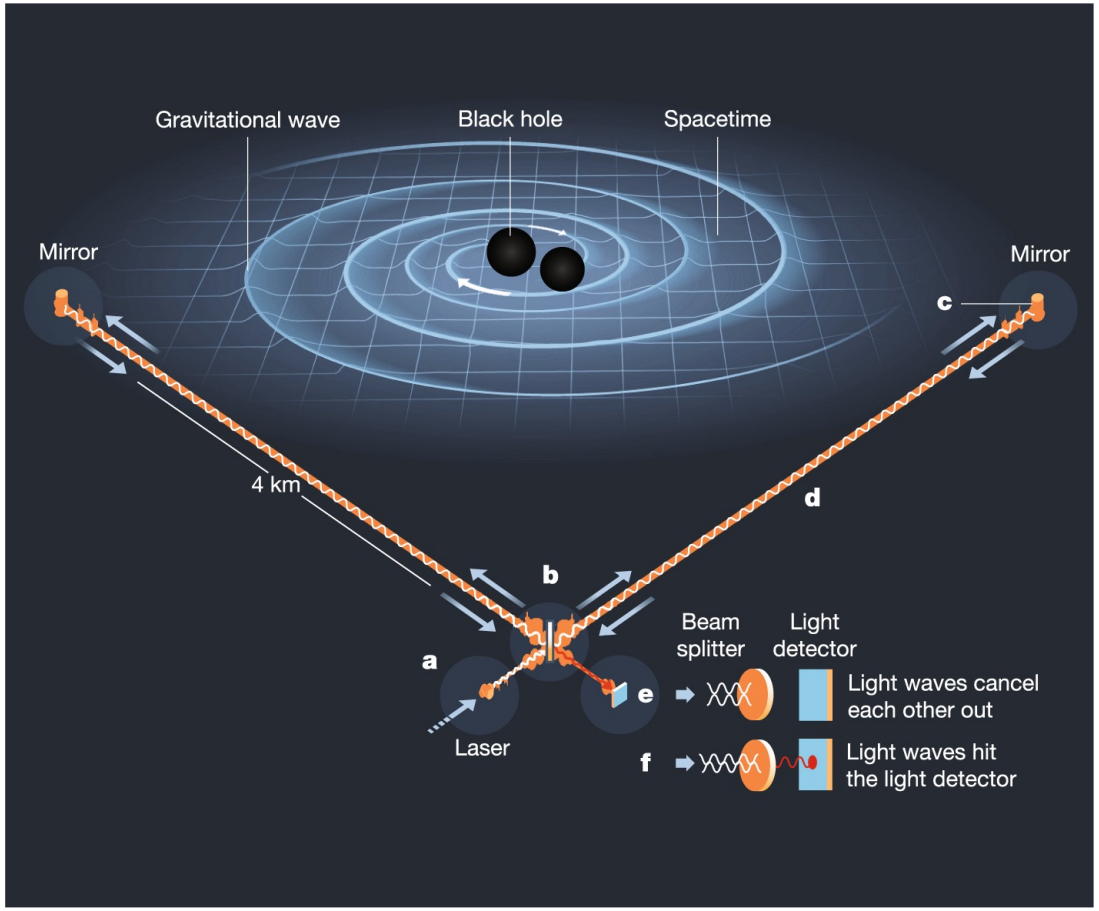


fluctuating phase shift

Squeezing reduces the noise without reducing the signal.

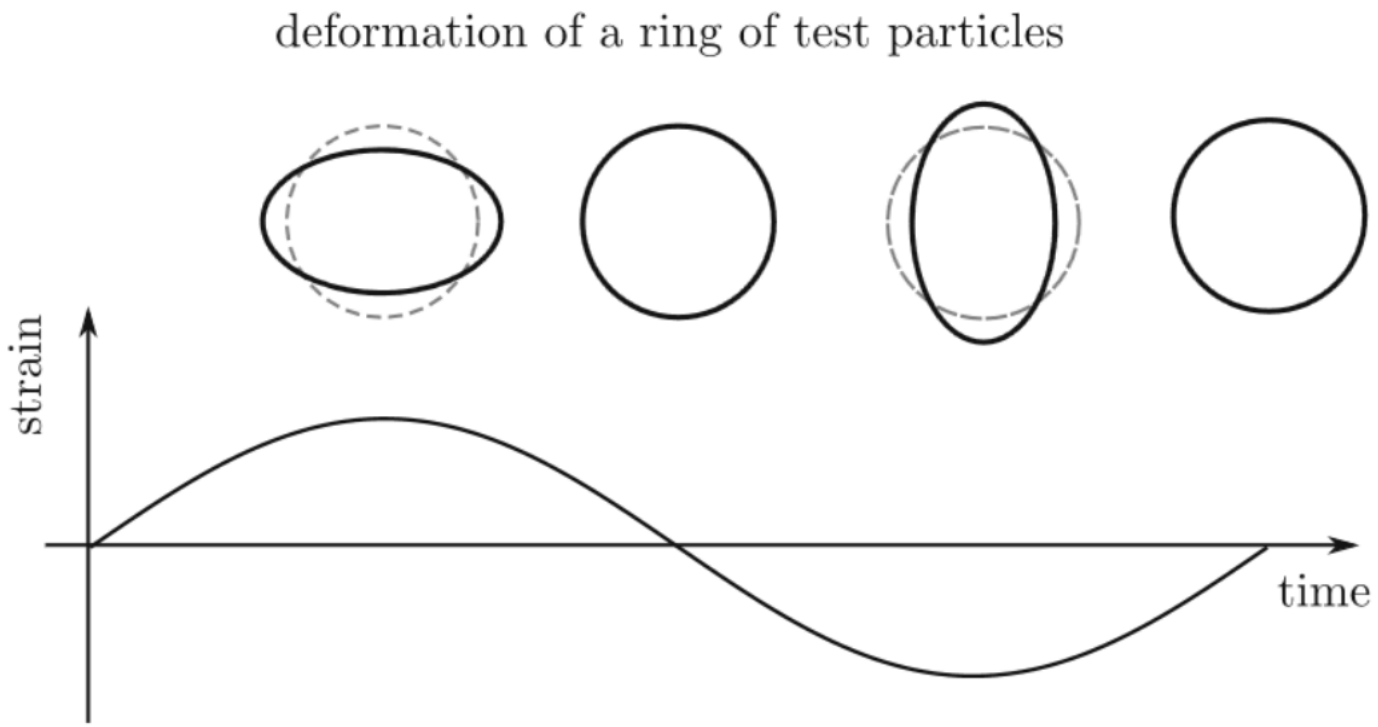
$$V^-_{sqz} < 1$$

Gravitational waves



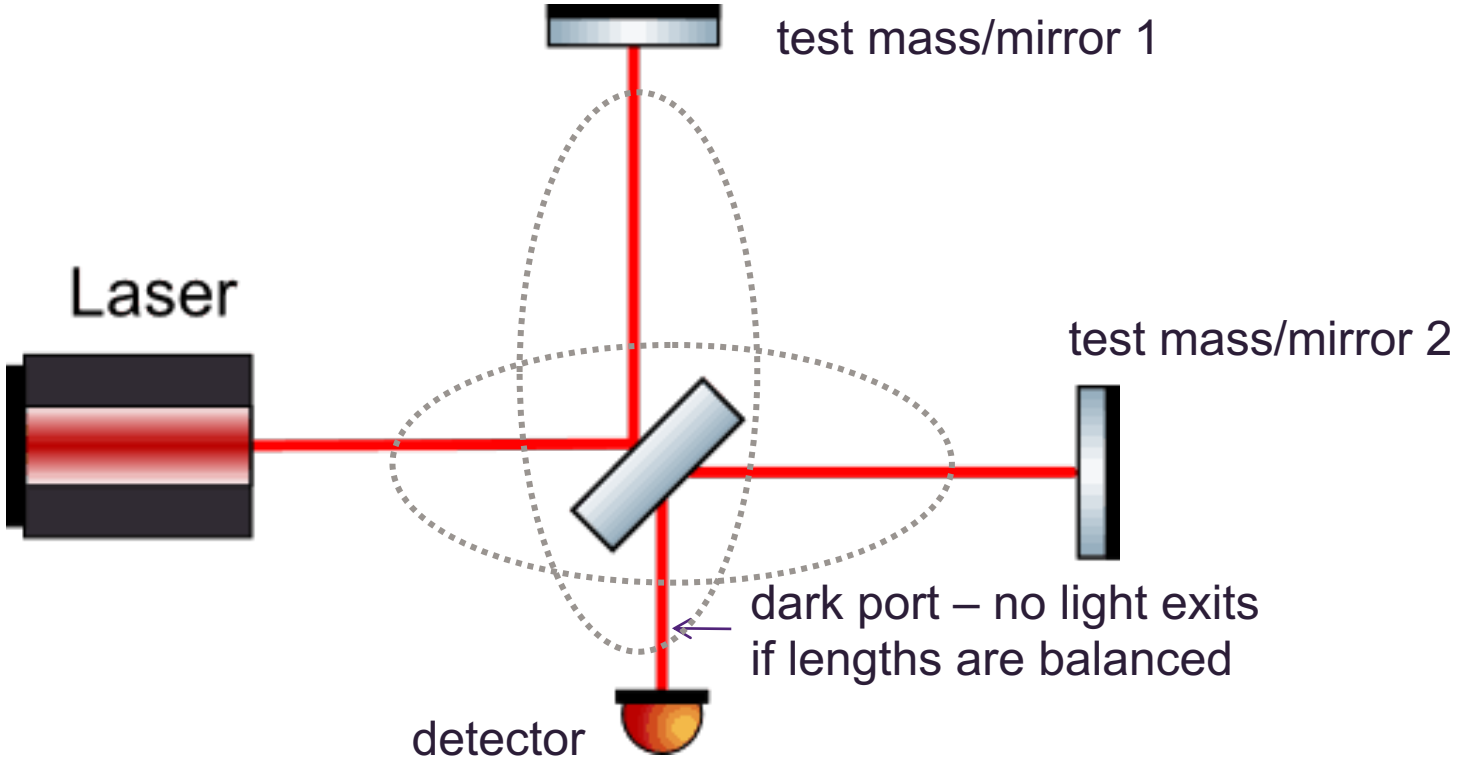
Miller, M.C., Yunes, N. The new frontier of gravitational waves. *Nature* **568**, 469–476 (2019).

Effect on test particles of gravitational waves



Bond, C., Brown, D., Freise, A. *et al.* Interferometer techniques for gravitational-wave detection. *Living Rev Relativ* **19**, 3 (2016).

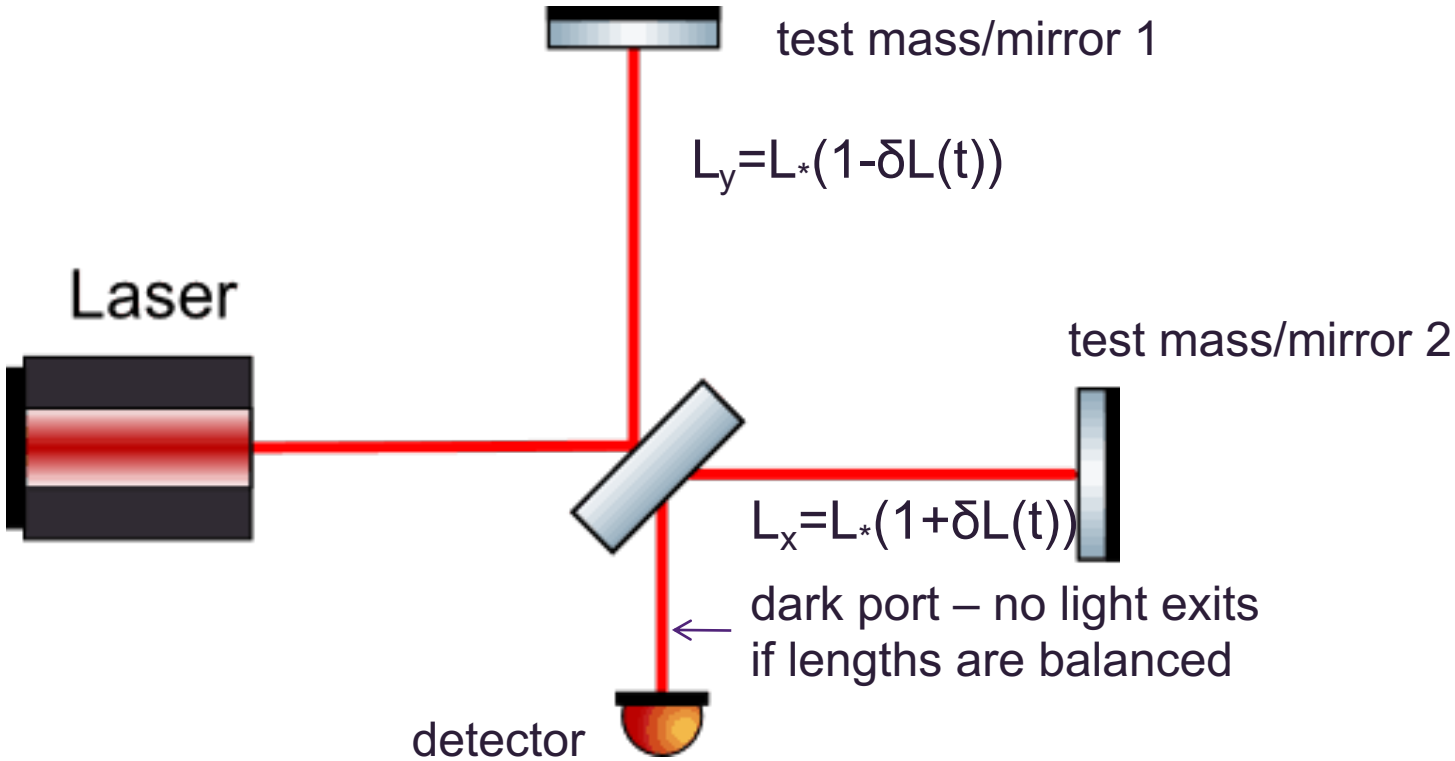
Gravitational wave interferometer



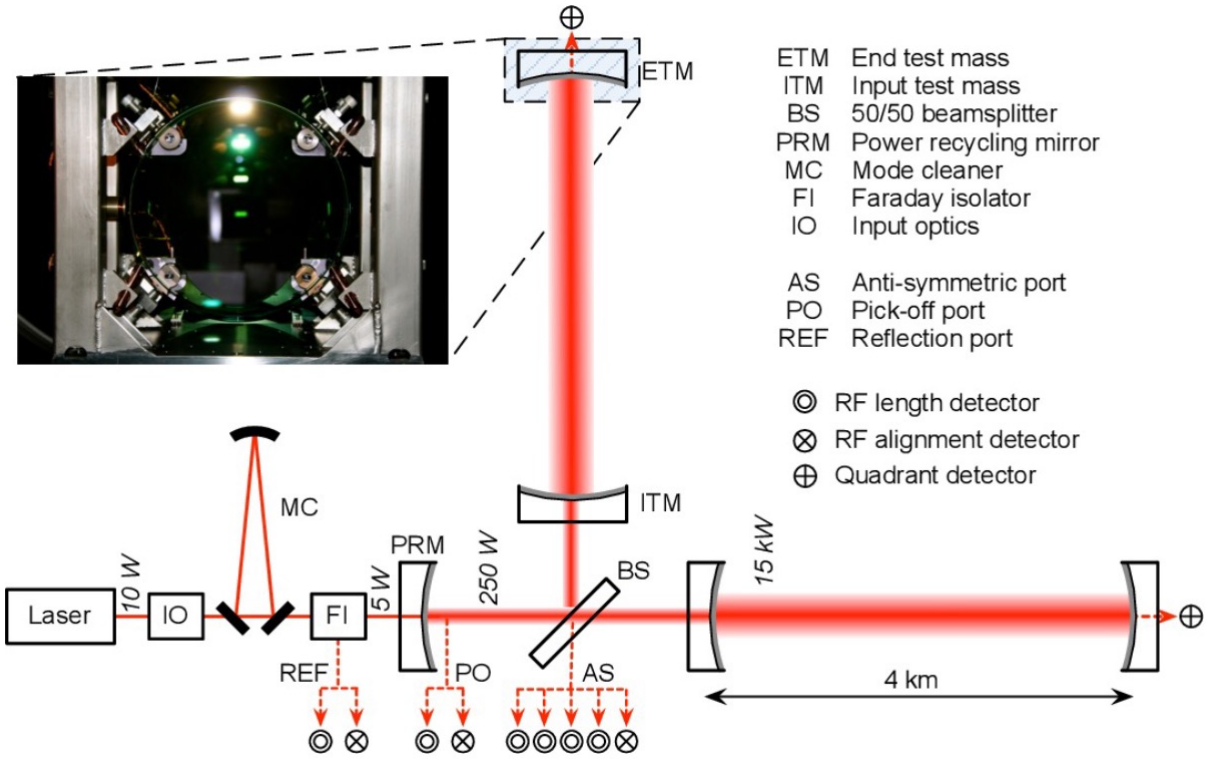
Gravitational-wave interferometer: scale of the effect

Dark port intensity $$\left|\alpha_d\right|^2 = \left|\alpha\right|^2 \omega^2 L_*^{\,2} a^2 \sin^2\left[\omega_g t\right]$$

For LIGO$|\alpha|^2$ = 200W, λ=1064nm, L_*= 4 km (X 75) leading to a sensitivity of 1: 10^{21}



Initial LIGO



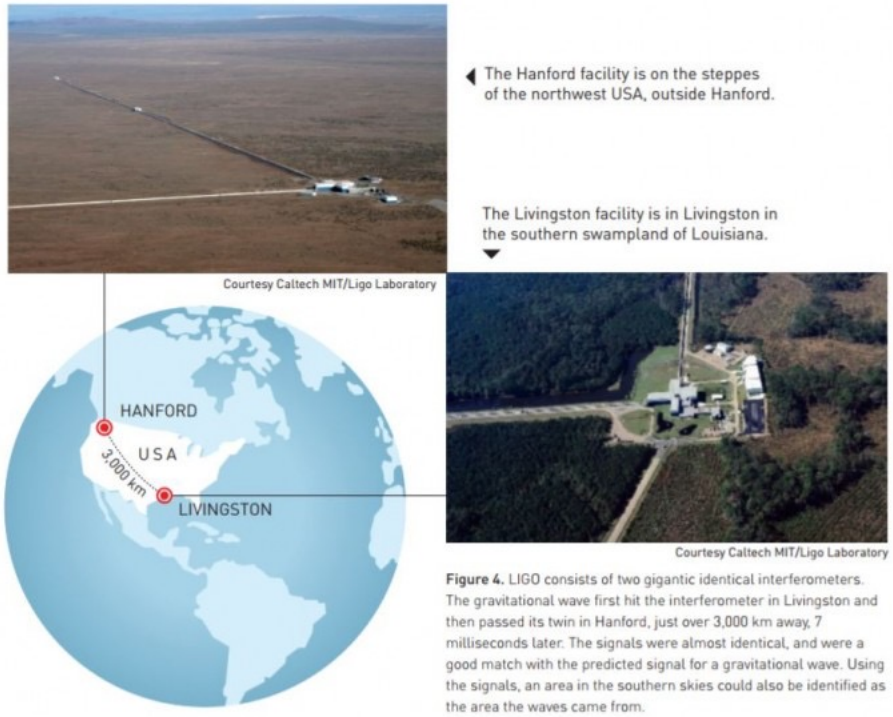
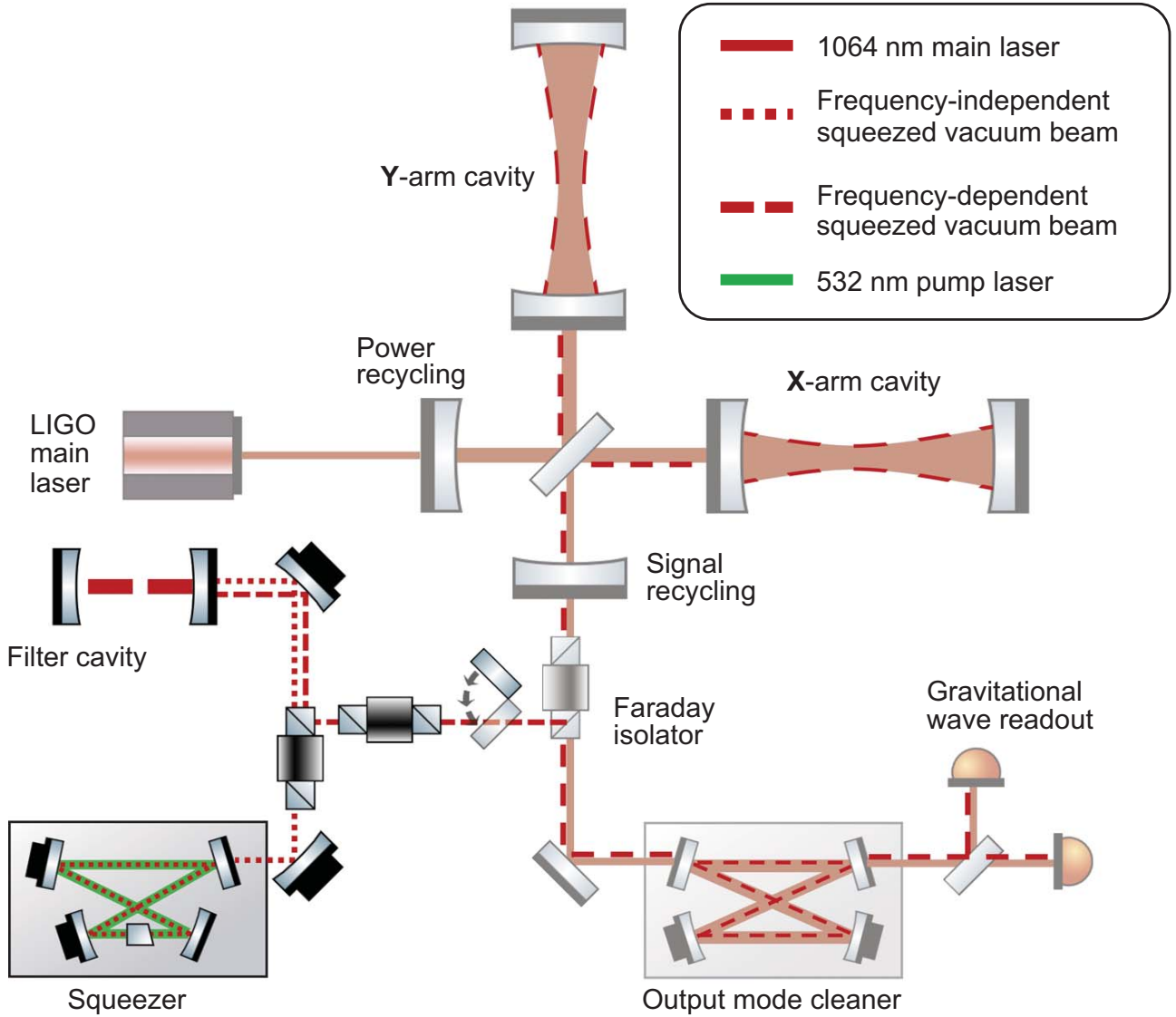
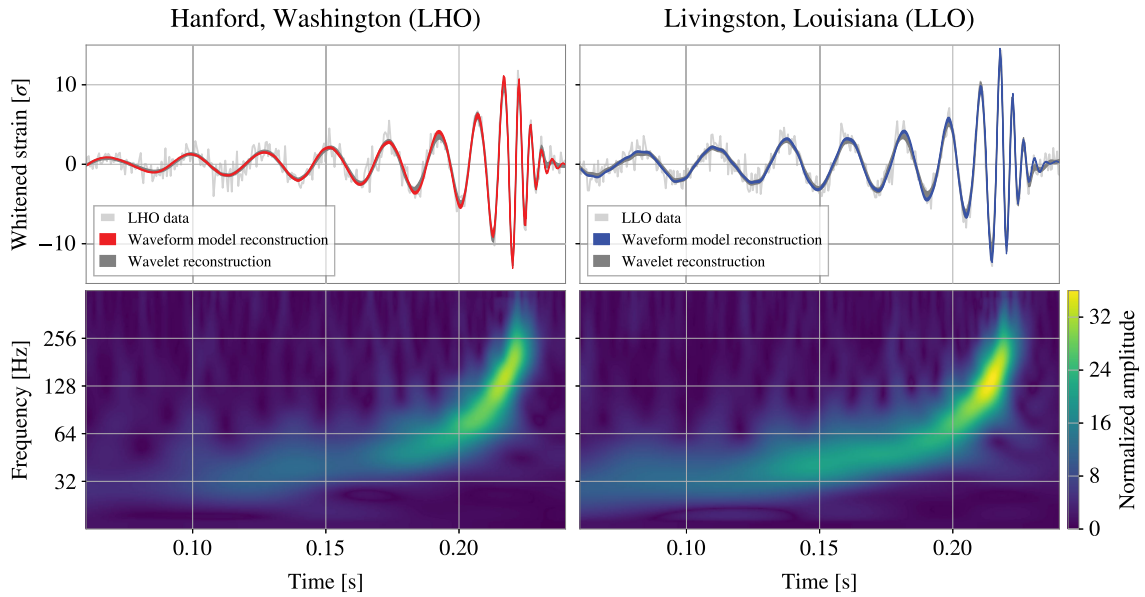


Figure 4. LIGO consists of two gigantic identical interferometers. The gravitational wave first hit the interferometer in Livingston and then passed its twin in Hanford, just over 3,000 km away, 7 milliseconds later. The signals were almost identical, and were a good match with the predicted signal for a gravitational wave. Using the signals, an area in the southern skies could also be identified as the area the waves came from.

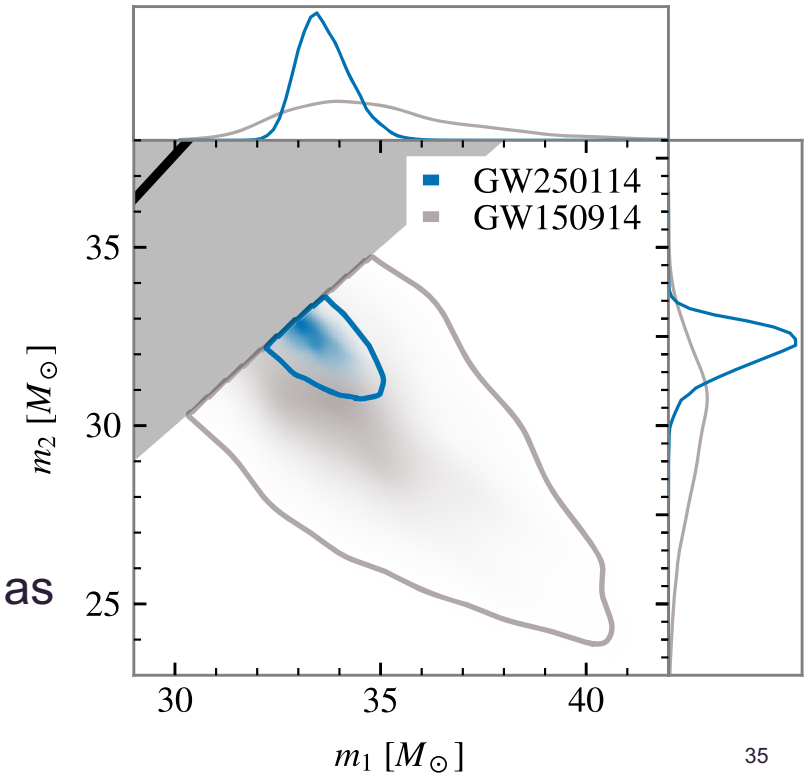
Advanced LIGO



2025 Gravitational wave detection by LIGO



Much higher SNR (thanks in part to squeezing) enables precision tests of the remnant black hole's ringdown



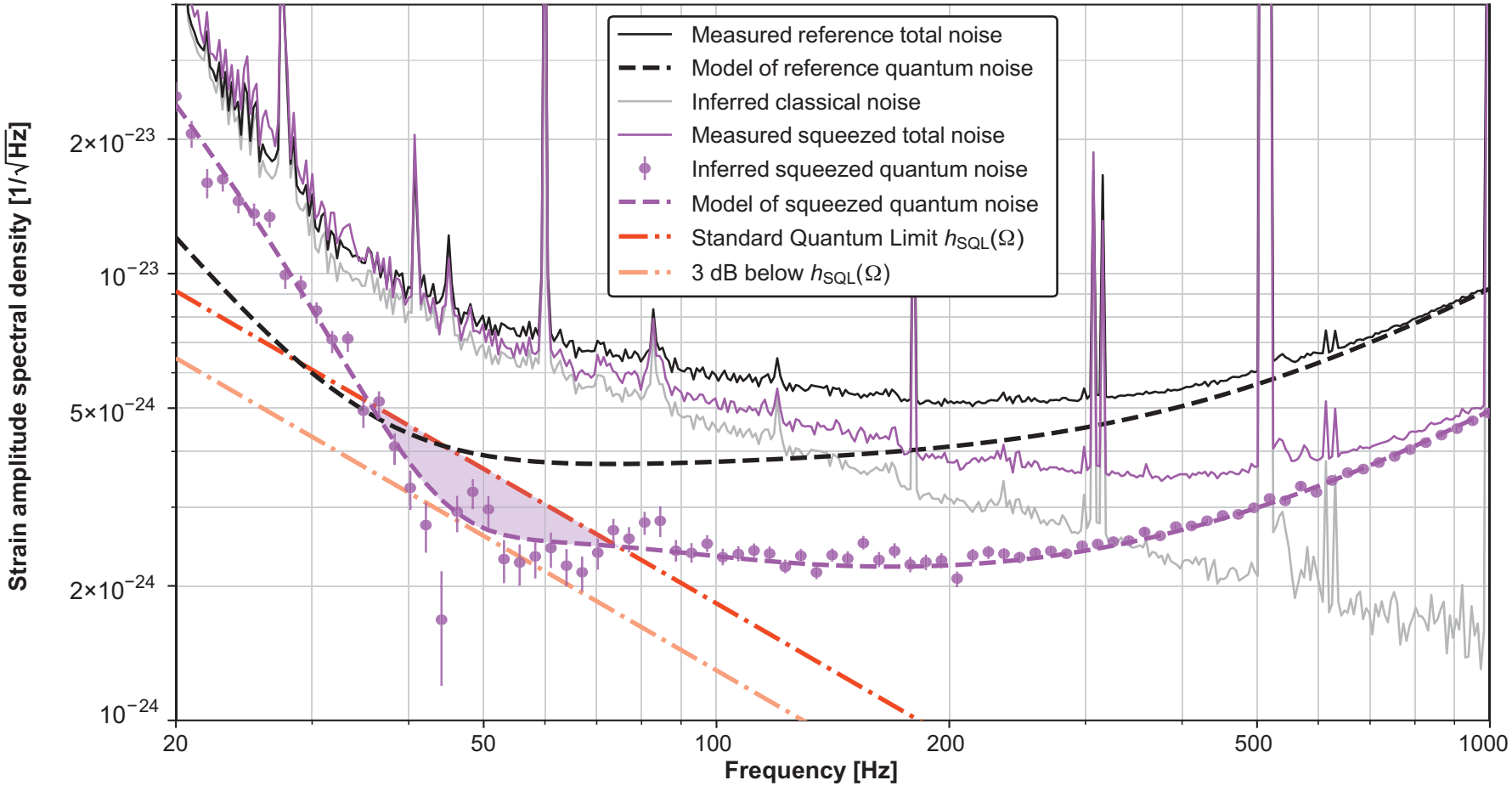
Black-hole masses, 33.6 & 32.2 solar masses;

1.3 Billion light years away;

Mass–energy equivalent of approximately 3.1 Suns radiated as gravitational waves.

Duration 230 ms

LIGO with Squeezing: measured noise



Summary

- **Fisher information determines precision:** classical Fisher information depends on the measurement, while quantum Fisher information gives the maximum information permitted by the quantum state.
- **Quantum states can improve phase sensitivity:** coherent light gives the standard quantum limit $\Delta\phi \propto \frac{1}{\sqrt{\bar{n}}}$, while optimised nonclassical probes can approach Heisenberg scaling $\Delta\phi \propto \frac{1}{\bar{n}}$.
- **Quantum metrology is used in gravitational-wave detection:** LIGO converts differential arm-length changes into a dark-port signal and uses squeezed vacuum to reduce quantum noise.